

Abdelrahman O. Sawy

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PROFILE

PhD student in Industrial & Systems Engineering at Rutgers University, specializing in **mathematical optimization**, **machine learning**, and **spatio-temporal analytics** for renewable energy systems and operations research. Founder of **The Agentic Lab**[®], an AI automation startup. Graduate teaching assistant across two courses; applied researcher with industry internship experience in automotive and manufacturing sectors.

EDUCATION

Rutgers University Expected January 2031
Ph.D. in Industrial and Systems Engineering — New Brunswick, NJ

Rutgers University January 2026
M.S. in Industrial and Systems Engineering, Concentration: System Analytics — New Brunswick, NJ **GPA: 4.00/4.00**

The American University in Cairo (AUC) June 2024
B.S. in Mechanical Engineering, Concentrations: Design & Industrial Engineering — New Cairo, Egypt

TECHNICAL SKILLS

Optimization & Operations Research: Gurobi, CPLEX, SciPy (LP/MILP/NLP), POSYDON framework, mathematical programming, deterministic & stochastic models, supply chain modeling

Machine Learning & Analytics: Python (scikit-learn, XGBoost, pandas, NumPy, Matplotlib), R / RStudio, predictive modeling, time-series forecasting, PCA, regression, MCMC, graph mining, missing data imputation

Programming: Python, MATLAB, R, LaTeX

Engineering Tools: ANSYS Fluent / Workbench, SolidWorks, CATIA

AI & Automation: Claude Code, LLM-based multi-agent workflow design, agentic system orchestration

RESEARCH EXPERIENCE

Graduate Research Assistant — Renewables & Industrial Analytics (RIA) Group Spring 2025 – Summer 2025
Rutgers University, New Brunswick, NJ Advisor: Dr. Ahmed Aziz Ezzat

- Developed AI-based energy forecasting models and spatio-temporal analytics pipelines for offshore wind farm operations & maintenance (O&M) optimization, targeting reductions in unplanned downtime.
- Simulated multi-turbine offshore wind farm O&M strategies using the POSYDON optimization framework; validated outputs against industry-standard platform Shoreline.io, producing statistically comparable performance metrics.
- Modeled predictive maintenance scheduling strategies integrating failure-rate estimates and weather data to minimize downtime and operational costs across renewable energy assets.

TEACHING EXPERIENCE

Graduate Teaching Assistant — Computational Methods in Industrial Engineering January 2026 – Present
Graduate Level · Rutgers University, New Brunswick, NJ

- Deliver graduate-level instruction on multivariate statistics, principal component analysis (PCA), regression, missing data imputation, recommendation systems, graph mining, Monte Carlo integration, and Markov Chain Monte Carlo (MCMC).
- Develop analytical derivations (eigenvalue decomposition, PCA optimization formulation) and Python/Excel-based computational templates; create machine learning implementation examples for 30+ graduate students.

Graduate Teaching Assistant — Probabilistic Models in Operations Research September 2025 – January 2026
Undergraduate Level · Rutgers University, New Brunswick, NJ

- Supported instruction in decision-theoretic models, Markov chain analysis, and queuing theory; developed Excel-based stochastic modeling instructional materials and guided students through implementation exercises.

PROFESSIONAL EXPERIENCE

Engineer Intern — MDA Service Sector, B.Tech

Winter 2023

Cairo, Egypt

- Analyzed performance datasets for major domestic appliances (MDAs), producing diagnostic reports that identified system inefficiencies and informed data-driven repair-cycle reduction strategies.
- Collaborated with service technicians to implement predictive maintenance workflows, decreasing unplanned service interventions and improving throughput.
- Maintained comprehensive service documentation ensuring data integrity and traceability for ongoing appliance performance tracking.

Engineer Intern — MTI Automotive Co.

Summer 2023

New Cairo, Egypt

- Identified workflow bottlenecks in engine, chassis, and bodywork repair operations through diagnostic software analysis and process mapping.
- Conducted time-and-motion studies; designed standardized operating procedures (SOPs) for key maintenance tasks, improving service consistency and reducing rework.
- Optimized parts supply-chain operations by analyzing inventory usage trends, forecasting demand, and applying just-in-time (JIT) principles to hybrid-vehicle battery inventory management.

PROJECTS & COMPETITIONS

Loan Default Prediction — Kaggle ML Competition

2025

- Ranked **21 out of 3,724 participants (top 0.6%)**; built and tuned XGBoost and stacked ensemble models with feature engineering to predict loan repayment outcomes.

Electricity Price Forecasting — P&G Forecasting Competition

2025

- Built electricity spot-price forecasting model leveraging Locational Marginal Pricing (LMP) data; applied time-series techniques (ARIMA, gradient boosting) and lag-feature engineering.

Depression Risk Classification — Kaggle ML Competition

2024

- Developed end-to-end ML pipeline with clinical feature engineering and ensemble classifiers for mental health risk prediction.

LEADERSHIP & ENTREPRENEURSHIP

Founder — The Agentic Lab® · theagenticlab.net

2025 – Present

- Founded AI automation startup delivering free agentic workflow services across multiple industry sectors; solutions and open tools are published at github.com/AOSawy and the company website.
- Developing a blog at theagenticlab.net covering trending developments in artificial intelligence and the global shift toward Artificial General Intelligence (AGI) and Artificial Superintelligence (ASI).

Marketing Head — AUC Developers Association

2022

- Led marketing strategy and outreach campaigns for the American University in Cairo's developer community; drove membership engagement and event attendance.

Additional: Member, AUC Mechanical Engineering Association — Racing & FabLab Teams (2024) · Student Recruiter, AUC CIMUN Model United Nations (2022) · Track & Field Team (2024)

RELEVANT COURSEWORK

Supply Chain Engineering · Engineering Decision Making Under Uncertainty · Forecasting Analytics · Data Analytics in Engineering Systems · Computational Methods in Industrial Engineering · Deterministic Models in Industrial Engineering · Quality Management · Data Mining · Planning & Operations Engineering · Production Analysis